

A Lloyd-Max-based Non-Uniform Quantization Scheme for Distributed Video Coding

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Abstract

In traditional video compression standards, the quantizer plays an important role. It is the same for the Distributed Video Coding (DVC), which can be thought to consist of a quantizer followed by a Slepian-Wolf encoder. The quantizer has dominant effect on the quality of the reconstructed video. In this context, this paper proposes quantization scheme for DVC based on a modified Lloyd-Max algorithm. The reconstruction values and partitions of quantizer are designed online for each Wyner-Ziv frame. The training set is extracted from the previous frame and a corresponding correlation model is computed respectively in the encoder and decoder. It is shown that at the cost of a slight higher error rate of the turbo code, the simple non-uniform quantization scheme will lower the quantization error in the reconstruction step, and on the whole improve the rate distortion performance.

1. Introduction

The traditional method for compression of video is to eliminate all redundancy at the encoder. This is a computationally intensive process, involving the motion estimation and motion compensation. At the same time, the decoder is far less complex than the encoder. However, with the proliferation of the mobile devices over the last decades many applications need the dual system, i.e., the low complexity encoders in power constraint environment and possibly at the expense of the high complexity decoders in a environment with rich resources.

An emerging approach to the low complexity encoding is distributed coding with side information. In the early 70's, Slepian and Wolf [1] showed that lossless source coding of two dependent i.i.d. signals X and Y is possible at $H(X, Y)$, even if encoding is done separately for X and Y , i.e., when the X -encoder does not have access to Y and

vice versa. A few years later, Wyner and Ziv formulated a rate distortion function for this scenario [2]. Research has shown that the concept of Wyner-Ziv coding can be applied to video compression as well, allowing a lighter encoder at the cost of a more computationally intensive decoder, which is also referred to as Distributed Video Coding (DVC). In [3] a simple pixel domain Wyner-Ziv video codec is proposed, in which all the sources statistics are exploited at the decoder following the Wyner-Ziv coding paradigm. However, it still suffers from a performance gap with respect to classical motion-compensated predictive coding solutions.

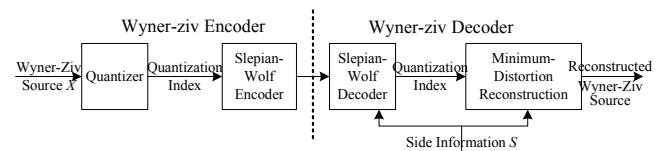


Figure 1. A WZ coder consisting of a quantizer and a SW encoder.

In general, a Wyner-Ziv codec can be thought to consist of a quantizer followed by a Slepian-Wolf encoder [7] as Fig.1 shows. In [3][4][8], all used in the Wyner-Ziv coding are uniform quantizers. [5] generalized the Lloyd algorithm for locally optimal fixed-rate Wyner-Ziv vector quantization. Unfortunately, vector quantizer dimensionality and entropy code block length are identical in their formulation, and thus the resulting quantizers either lack in performance or are prohibitively complex [7]. [6] proposed a rate measure that depends on both the quantization index and the side information, which divorces the dimensionality of the quantizer from the block length of the Slepian-Wolf coder. However, in this quantization scheme the correlation model between the side information and the Wyner-Ziv frames should be estimated in advance. It means the correlation model of the video sequences should be estimated off-line, and then it is not suitable for a practical DVC codec.

In this context, this paper proposed a simple and practical frame-level non-uniform quantization scheme for distributed video codec. We utilize a modified Lloyd-Max algorithm to design a quantizer both in the encoder and in the decoder. On account of the current frame is not available in the decoder, the training set data is extracted from the previous frame $X(2t - 1)$, which is encoded by H.264 intra mode, and in the decoder has been decoded successfully. So the codebook and the partition used for decoding the current frame $X(2t)$ can keep consistency with the quantizer in the encoding. The corresponding correlation model between the current frame $X(2t)$ and the previous frame $X(2t - 1)$ is computed and used for quantization in the encoder. For the same reason, the accurate correlation model is not known for the quantizer in the decoder, so we replace it with the one between the previous decoded key frame and Wyner-Ziv frame. Although in the modified Lloyd-Max algorithm at the decoder, the training set data is simply replaced by the decoded previous frame and an inaccurate correlation model is used, the proposed simple non-uniform quantization scheme can improve the rate distortion performance while introducing a slight higher turbo code error rate compared with the uniform scheme.

This paper is organized as follows: Section 2 presents a brief summary of the codec architecture. In section 3, the frame level non-uniform quantization scheme for distributed video codec is described. Several experiments are performed, in Section 4, to evaluated and compare the coding efficiency of the proposed approach and, finally, in Section 5, conclusions and some future work topics are presented.

2. The distributed video codec architecture

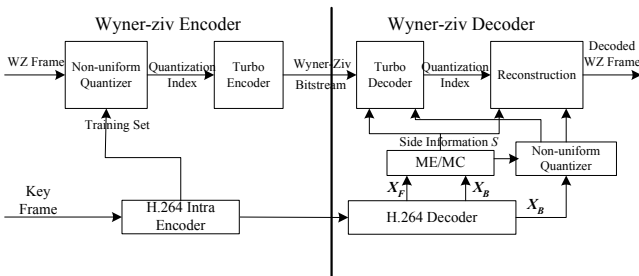


Figure 2. The Wyner-Ziv codec architecture.

The system we consider in our work is essentially an implementation of the system described by [3]. A general overview of the system is given in Fig. 2. It illustrates the architecture of the Wyner-Ziv codec including the proposed frame level non-uniform quantizer, which is different from the uniform quantizer of [3]. In the encoder, selected key

frames (often every second frame of the video sequence) are encoded using intra coding mode of the well-established video compression standards, such as H.263, H.264, etc. The intermediate frames are quantized and encoded following the Wyner-Ziv principles. The decoder first decodes the key frames, and then reconstructs intermediate frames by motion compensated temporal interpolation between adjacent key frames. The main blocks of the Wyner-Ziv video codec are discussed in more detail below.

2.1. Wyner-Ziv Encoder

First, the input video frames are divided into two categories. Each frame has a length of N pixels, each pixel is considered as one symbol. The odd frames are the key frame and encoded by intra coding, here we used is H.264 codec in intra mode. The even frames are called Wyner-Ziv frames, in which each symbol is quantized to an index using

$$F_q : x^N \rightarrow u^N \quad (1)$$

with $U = \{0, \dots, 2^L - 1\}$. In our proposed system, F_q implements a non-uniform scalar quantizer with $2^L = 16$ levels. The quantization algorithm based on a modified Lloyd-Max algorithm will be described in section 3 in detail. In order to fulfill the algorithm, the training set data should be extracted from the previous frame, which refers to the H.264 intra encoded frame. The resulting quantizer index U is then encoded using a systematic turbo code with two identical 16-state convolutional constitute codes with rate $4/5$, which is the same as in [3], more details please refer to [9][10]. Note that due to the choice of constituent codes, the rate of the turbo code in our system can never be larger than 2 bits per symbol, which means 0.5 bits/bit. The output of the turbo code can be punctured to a specific lower rate R (bits/symbol), yielding the code of length RN bits. The turbo encoder, including puncturing, is represented by the following equation:

$$F_e : u^N \rightarrow c^{RN} \quad (2)$$

typically with $C = \{0, 1\}$.

2.2. Wyner-Ziv Decoder

At the decoder, the ME/MC module is used to generate the side information frame, an estimation of the current Wyner-Ziv frame, based on the previously decoded frames, X_B and X_F . In this paper, X_B and X_F refer to the previous and the next temporally adjacent key frames. Side information frame is given by (3), where $X_B(x + dx_b, y + dy_b)$ and $X_F(x + dx_f, y + dy_f)$ represent the backward and the forward motion compensated frames, respectively, and (x, y) corresponds to the pixel location in the side information

frame; (dx_b, dy_b) and (dx_f, dy_f) represent the motion vectors for X_B and X_F , respectively.

$$S(x, y) = \frac{1}{2}(X_B(x+dx_b, y+dy_b) + X_F(x+dx_f, y+dy_f)) \quad (3)$$

After side information frame is got, the quantization index is computed by the inverse quantizing step

$$F_{iq} : c^{RN} \times s^N \rightarrow \hat{u}^N \quad (4)$$

(4) yields the decoded quantization index $\hat{U}$, which maybe unequal to U due to decoding error. And then a posteriori probability soft input soft output algorithm is used to iteratively computing the reconstructed input source based on

$$F_d : \hat{u}^N \times s^N \rightarrow \hat{x}^N \quad (5)$$

In this process the Laplacian distribution parameter is computed in frame level as [8] describes. Because we use the non-uniform quantizer in the encoder, the optimal reconstruction in this step can be defined as

$$\begin{aligned} F_d : \hat{x}^*(\hat{u}, s) &= E[X | Q = \hat{u}, S = s] \\ &= \frac{\sum_{x:Q(x)=\hat{u}} x \cdot p_{X|S}(x | s)}{\sum_{x:Q(x)=\hat{u}} p_{X|S}(x | s)} \end{aligned} \quad (6)$$

where $P(X|S)$ means the correlation model between the source frame and the side information frame.

3. The frame level non-uniform quantization scheme

Just as mentioned in the introduction, the described system is actually a two-tier system consisting of an inner Slepian-Wolf codec for lossless compression of a quantized version of the input data, followed by an estimator, which essentially bounds and improves the error in the side-information by a conditional average [9]. The minimal rate, and thus the maximum compression ratio, is solely determined by $H(U|S)$. The transmission rate R at which error-free construction is theoretically possible, is bounded by the Slepian-Wolf theorem[1]:

$$\begin{aligned} R &\geq H(U | S) \\ &= H(U | S) \\ &= \sum_s P(s) H(U | s) \\ &= \sum_s P(s) \sum_u P(u | s) \log P(u | s) \end{aligned} \quad (7)$$

where $P(u|s)$ can be calculated from $P(x|s)$ and the quantization rule as following

$$\begin{aligned} P(u | s) &= P(u | x) \cdot P(x | s) \\ &= \sum_{x:Q(x)=u} P(x | s) \end{aligned} \quad (8)$$

This conditional entropy is largely dependent on the design of scalar quantizer F_q . So in distributed video coding, because of the inequality of the source symbol, a uniform quantizer yields a lower conditional entropy $H(U|S)$ compared with non-uniform quantizer according to information theory, and therefore higher compression ratio, but also a larger quantization distortion after F_q . This means that lowering the error rate of the turbo code increases the possible quantization error in the reconstruction step. On the contrary, we can lower the quantization error in the reconstruction step by slightly increasing the error rate of the turbo code. The experiments results in section 4 prove the idea reasonable.

From (6), the function F_d , assuming the turbo codec had no decoding errors, will minimize the error by taking the mean of the bin weighted by the probability conditioned on the side-information and the decoded quantizer bin. So according to (6) we propose a modified Lloyd-Max algorithm to design a quantizer to quantize source X to produce the index U in the encoder, and compute the $\hat{U}$ by the same algorithm in the decoder. This is a compromise, which can lower the quantization error in the reconstruction step, and at the same time just introduce a slight error rate of the turbo code that will not be enough to deteriorate the quality of reconstruction.

Now the remain question is, the side information cannot be accessed by the encoder if we use (6) to design a quantizer. The temporal correlation between the video frames supply one replacement. In fact in the decoder the side information is not accurately known, just estimated by the previous decoded frame. In [8] the frame level Laplacian distribution parameter α of the correlation model for decoding is computed according to X_B and X_F frames. Following the same principles, in the decoder we utilize the previous decoded key frame and Wyner-Ziv frame to compute a frame level Laplacian distribution parameter α , while in the encoder we utilize the frame $X(2t-1)$ to estimate the correlation model.

The modified Lloyd-Max algorithm in the encoder is described as following:

For each Wyner-Ziv frame X_{2t} :

1. Calculate the Laplacian distribution parameter α between $X(2t-1)$ and $X(2t)$;
2. Select training set data from $X(2t-1)$, general $1/32$ of the $X(2t-1)$;
3. $k = 1, last_distor = 0, tol = 0.0000001$;

4. Choose the uniform quantizer as the initial quantization function $q_0(x)$;
5. Obtain the optimal reconstruction function $x(u, X_{2t-1})$ by (6), for the previous quantization function $q_{k-1}(x)$;
6. Obtain the optimal quantization function $q_k(x)$ for the reconstruction function $x(u, X_{2t-1})$;
7. Compute the distortion $distor$;
8. If $\text{abs}(distor - \text{last_distor})/distor > \text{tol}$, then $\text{last_distor} = distor$, $k++$, and goback to 5.

end

The quantizer used in decoder also utilizes above algorithm, except that α is estimated based on the previous and next decoded key frames as [8] describes. The difference is owing to that in DVC paradigm the encoder cannot access the side information, at the same time the decoder cannot access the current frame. Although The difference between the quantizing algorithm in the encoder and decoder leads a little degradation in the rate distortion performance, it is lucky that on account of the high temporal correlation between successive video frames the quantization scheme still works efficiently, which is verified by the following experiment results.

4. Experiments Results

To evaluate the efficiency of the proposed scheme, several experiments have been performed. In the simulations, the Wyner-Ziv frame bit rates keep constant, the request more parity bits function in [3] are not used. In ME/MC the search range is 16, and the block size is 16×16 . The quantization levels are 16. The SISO APP decoding runs at most 16 iterations. When the PSNR of current iteration has no change compared with last iteration's, the decoding of current frame is considered to be successful and will be broken; otherwise decoding runs until 16 iterations reaching. The key frames are encoded by H.264 intra mode, the PSNR of the decoded Y component of the key frames is 37.31 dBs in average, and the bit rate is 925.29 kbit/s @ 30.00 Hz.

Fig.3 and Fig.4 show the rate distortion (RD) results for the first 100 frames of *Foreman* QCIF video sequences coded by the uniform and non-uniform quantization schemes respectively, and in turbo code rate of 0.5 bits/bit and 0.4 bits/bit respectively. Fig.3 shows that the non-uniform quantizer allows achieving better RD performance compared to the uniform one. The improvement is from 0.1dBs to 0.6dBs, on average 0.36dBs. This suggests that if the side information used in two scheme remains same, the non-uniform quantization scheme supply a

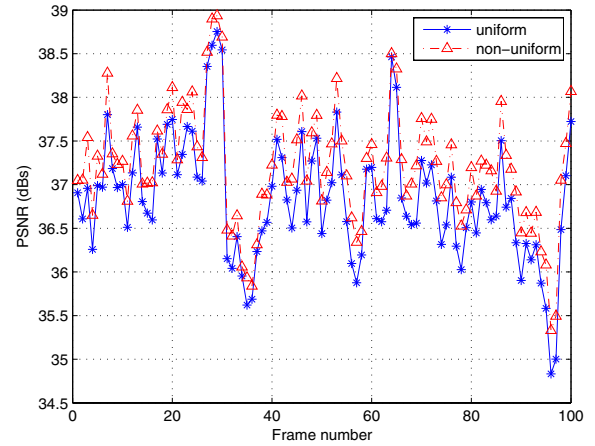


Figure 3. RD performance of every WZ frame in turbo code rate of 0.5 bits/bit.

potential way to achieve higher reconstruction quality. The same performance can be observed in Fig.4. However, Fig.4 shows that some frames can be decoded successfully under the uniform scheme, but unsuccessfully under the non-uniform scheme, e.g., 57 and 90 frames. The reason will be explained through the analysis of Fig.5 and Fig.6 below.

Fig.5 and Fig.6 show the average RD performance and symbol error rate of the first 100 frames of *Foreman* in varied turbo code rate varied from 0.2 to 0.5bits/bit. From Fig.5 it can be noticed when the turbo code bit rates vary from 0.5 to 0.3bits/bit, the PSNR of non-uniform scheme decreases higher than the uniform scheme. Fig.6 shows that the lower the turbo code bit rates, the higher the symbol error rate of the non-uniform scheme in comparison with the uniform scheme. In other words, the non-uniform quantization scheme need more turbo code bits to correct symbol errors than uniform quantization. This is in accordance with the analysis in section 3. For this reason some frames which are decoded unsuccessfully under the non-uniform scheme can decode successfully under the uniform scheme, which is in accordance with Fig.4. However, observing Fig.5 and Fig.6 jointly, we find the non-uniform quantization scheme can get higher reconstruction quality in the expense of a slight higher symbol error rates. In the turbo code rate of 0.5bits/bit both symbol error rates of the two quantization scheme are zero, the non-uniform scheme possess a higher RD performance than the uniform one. Although in the turbo code rate of 0.4bits/bit the symbol error rate of the non-uniform scheme is more than the uniform scheme about 0.015, which almost owes to the frames decoded unsuccessfully, its reconstruction quality is still higher than the uniform one about 0.05dBs on average. This suggests that at

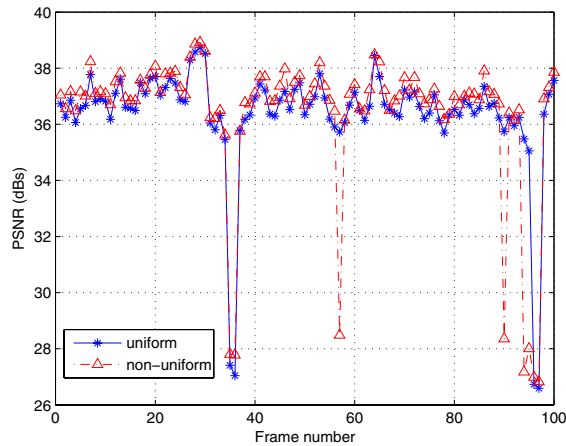


Figure 4. RD performance of every WZ frame in turbo code rate of 0.4 bits/bit.

the cost of a slight higher error rates of turbo decoding, the non-uniform scheme will lower the quantization error in the reconstruction step, and on the whole improve the rate distortion performance.

5. Conclusion

In this paper a non-uniform scheme for DVC has been considered. In this context a modified Lloyd-Max algorithm has been proposed and a practical non-uniform scheme are explored. From the experiment results and analysis, it is found in the DVC system the non-uniform quantization scheme supply a potential way to achieve higher reconstruction quality. At the expense of a slight symbol error rate of the turbo code, the non-uniform scheme will lower the quantization error in the reconstruction step, and on the whole improve the rate distortion performance. As future work, it is planned to formalize the RD performance of the non-uniform quantizer for DVC, and the joint quantization scheme of both key frames and Wyner-Ziv frames will be explored.

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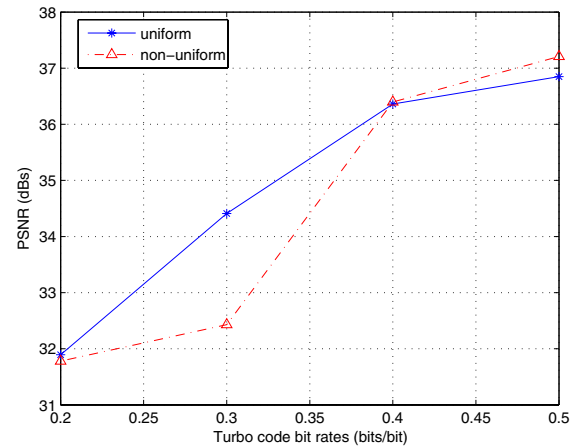


Figure 5. Average RD performance

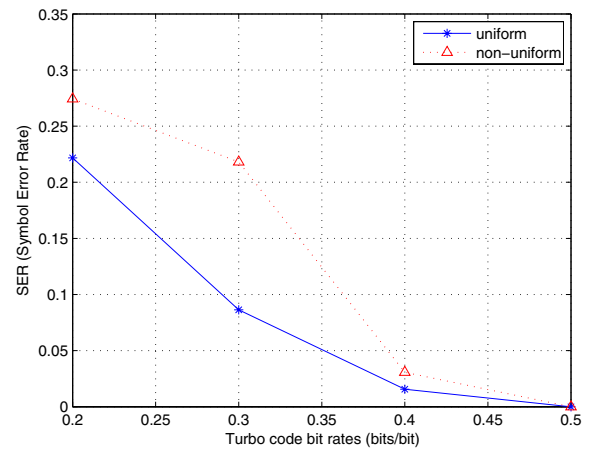


Figure 6. Average symbol error rate.

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